Experimental Validation of a Corrugated Membrane for Large-Scale Liquid Hydrogen Cargo Containment

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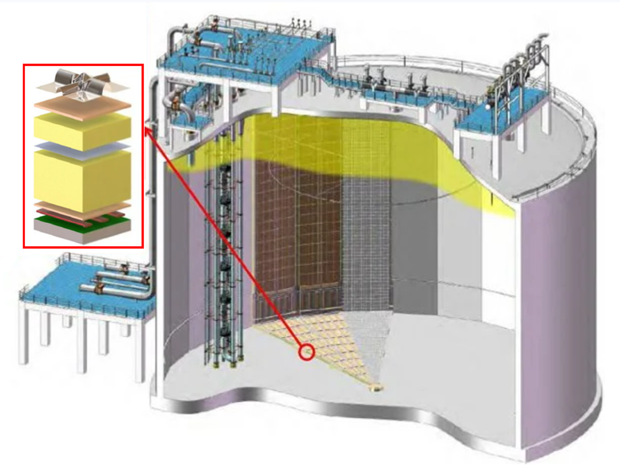
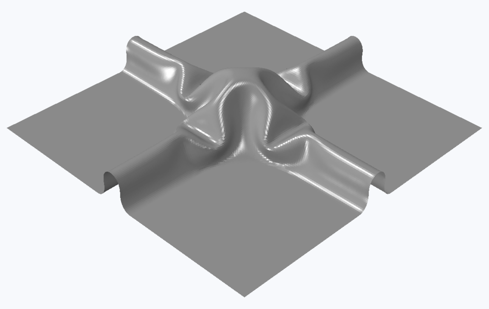
Introduction

Large-scale liquid hydrogen (LH_2) transport requires thin metallic containment barriers that remain gas-tight while accommodating thermal contraction and restrained in-plane deformation. Corrugated 316L stainless-steel membranes offer a promising solution because their geometry can redistribute imposed strain and reduce local deformation demand in nominally planar regions. Although the target application is cryogenic, room-temperature mechanical testing is used here to isolate the geometry-driven deformation-accommodation mechanism under controlled displacement loading, rather than to reproduce the full LH_2 service environment. Cryogenic integrity is examined separately through liquid-nitrogen (LN_2) cycling of a 0.2 m^3 integrated demonstrator, while full mechanical qualification at LH_2 temperature remains a subsequent validation step.

This study experimentally evaluates the corrugated membrane concept using full-field digital image correlation (DIC) under tensile loading, acoustic-emission-monitored cyclic testing, and LN_2 thermal cycling with pressure-retention and leak-check assessment. The results clarify how corrugation features accommodate deformation and provide preliminary evidence for assembly-level integrity, supporting further development of corrugated membrane systems for large-scale LH_2 cargo containment.



Membrane-type LH_2 storage-tank concept and layered containment barrier.



Corrugated membrane geometry used to absorb restrained in-plane contraction.

(C) Experimental validation route

Room-temperature DIC tensile test
Controlled displacement loading

Isolate corrugation-driven strain redistribution
Mechanism test, not full LH_2 qualification

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AE-monitored cyclic loading
Target 400-cycle window

Screen cyclic activity and deformation stability
Damage monitoring, not standalone crack proof

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0.2 m^3 integrated demonstrator
5× LN_2 filling/draining cycles

Apply cryogenic thermal excursions at assembly scale

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Pressure hold + helium leak check
Gas-tightness assessment after cycling

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Present evidence
Deformation accommodation + cyclic-response screening
+ preliminary cryogenic integrity

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Next qualification step
Mechanical validation at LH_2 temperature

Staged validation route linking room-temperature mechanism tests to LN_2 cycling and future LH_2 qualification.